Medlyn: Task-Specific Clinical Views from Located Evidence

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Medlyn project

Technical report: experimental prototype

Abstract

We present Medlyn, a prototype that constructs a source-linked clinical view from German documents for a clinician's role and task. We extract located facts, select fact identifiers and render registered templates through a shared claim validator. In an agent-reviewed API check of the challenge situations, each primary cold matrix answered 15 of 15 questions, with 166 of 166 cited facts located and 0 unsupported lines found among 167 reviewed lines. The slowest primary cold request took 10.138 s, missing the under-10 s target; confirmed cached requests took at most 0.239 s. On synthetic held-out claims, our local verifier reached macro F-score 0.764, compared with 0.896 for GPT and 0.295 for rules. An offline cascade reached 0.956 with a projected 55% fewer GPT calls. On the separate real-text rule-labelled set, the cascade trailed GPT. These results are agent-reviewed, not clinician-reviewed, and do not establish clinical efficacy.

1 Introduction

A longitudinal patient record contains more information than a clinician needs for a single task. The relevant subset changes with the task. Preparing an intervention, reviewing a tumour-board recommendation and checking follow-up require different facts from the same documents. An accurate general summary can omit a required answer. A fluent task-specific summary can also change a recommendation into a completed procedure or treat missing documentation as a negative finding.

We study the construction of a clinical view with explicit evidence constraints. Medlyn maps a role and task to information needs, selects stored facts and presents short English statements linked to German passages. We call this view a *lens*. We use language models for extraction and selection, but registered templates render the fact lines. When a model composes a headline or answer, we validate its claims against the cited facts and original context before release.

Our contribution is a system design that joins task-specific selection to a common claim boundary. We use the same patient, corpus revision and evidence window across retrieval, documentation checks, findings and letters. We keep availability, assertion, temporal status, completeness, conflict and clinician review as separate state dimensions. We further describe a synthetic data pipeline and a local verifier experiment. These components provide testable mechanisms; their implementation alone does not establish their accuracy.

We use transformer language models (Vaswani et al., 2017) for extraction and selection, with source evidence retained outside the model parameters. We organise the report around the architecture, data construction and verification experiment. We exclude challenge passages, patient-level predictions and private evaluation evidence from this report.

2 Background

We distinguish factual support from answer coverage. Factuality studies in summarisation show why fluent text needs evaluation against its source (Maynez et al., 2020; Pagnoni et al., 2021). In clinical summarisation, both the evaluation metric and the kind of error matter. Adams et al. (2023) examine faithfulness metrics for hospital-course summaries, and Zhang et al. (2023) study medical-summary faithfulness. The reader study of Van Veen et al. (2024) evaluates adapted models on specified summarisation tasks. We do not transfer those results to our workflow or use a generic text score as a measure of clinical utility.

Evidence attribution and atomic decomposition provide relevant design choices. ALCE evaluates answer quality and citation support (Gao et al., 2023b); RARR retrieves attribution and revises unsupported output (Gao et al., 2023a). FActScore assesses the support of atomic facts (Min et al., 2023). FactsR uses salient facts as an intermediate representation for healthcare documentation (Hansen et al., 2025). We adopt the separation of evidence and presentation, then constrain the final fact lines with templates. TBFact, described with the Healthcare Agent Orchestrator (Blondeel et al., 2025), motivates measuring additions and omissions separately. We do not claim to reproduce its benchmark.

A fact can be located in a document while its interpretation is wrong. NegEx (Chapman et al., 2001), ConText (Harkema et al., 2009) and German negation detection (Cotik et al., 2016) motivate explicit assertion and temporal labels. SummaC (Laban et al., 2022), AlignScore (Zha et al., 2023) and MiniCheck (Tang et al., 2024) provide approaches to checking consistency between claims and evidence. We retain location and semantic support as separate checks. We also distinguish strict contradictions from contextual ambiguity, following the problem formulation studied by Lu et al. (2026).

Retrieval-augmented generation combines retrieved evidence with generation (Lewis et al., 2020). Dense passage retrieval (Karpukhin et al., 2020) and lexical ranking address different matching failures. We combine lexical and vector candidates while reserving absence claims for a full-record check. For synthetic data, Synthea (Walonoski et al., 2018), Asclepius (Kweon et al., 2023) and longitudinal note generation (Poulett et al., 2026) motivate separating patient histories from rendered notes. We require labels to survive in the final text and distinguish code checks from human annotation.

3 System Architecture

Figure 1 gives the main data and control paths. We ingest evidence before serving a lens. The live path selects facts, checks apparent documentation gaps and validates claims. We store clinician reviews in an append-only overlay. Accepting a statement records a review action; it does not increase the statement's support in the documents.

3.1 Ingestion and Location

We normalise line breaks and encoded entities while retaining a map to raw offsets. We segment the normalised text into headings, sentences, list items and medication rows. The segmenter protects common German abbreviations and joins wrapped prose. Structured extraction returns a fact kind, an English label, attributes, assertion, temporal status, event date and a source quote. We locate the quote within its segment and check sensitive values, including numbers, doses, dates and sides. We drop failures rather than store an unlocated fact.

We then compare the extracted meaning with the passage and neighbouring context. A repaired fact must pass location again. A negation trigger can request another semantic check; it does not decide the clinical assertion. We retain the source date and document identity on medication lists. We do not describe a historical list as current medication. LangExtract (Goel, 2026) provides a related source-grounded extraction tool, but its output would still require an independent label check.

For page-based inputs, we retain extracted text together with page and bounding-box provenance. Text recognition errors can produce apparently valid offsets into an incorrect transcription. We therefore distinguish text location from agreement with the rendered page. Image embeddings can retrieve a page; they cannot establish the support of a clinical statement. End-to-end PDF and scan accuracy remains an evaluation requirement.

3.2 Retrieval

We define knowledge-time eligibility once in the store. For patient p, corpus revision r and as-of date t, the eligible document set is

$$E(p, r, t) = \{d : \text{patient}(d) = p,\ \text{revision}(d) = r,\ \text{docDate}(d) \leq t\}. \tag{1}$$

All consumers receive this set. An eligible document can describe a future procedure; we retain its planned status rather than filtering on the event date. Revision guards reject reads whose evidence identity changed during the request.

For the small challenge record, the core selector receives every eligible fact of the relevant kinds. Investigation search and the scale extension use BM25 (Robertson and Zaragoza, 2009) and Gemini embeddings

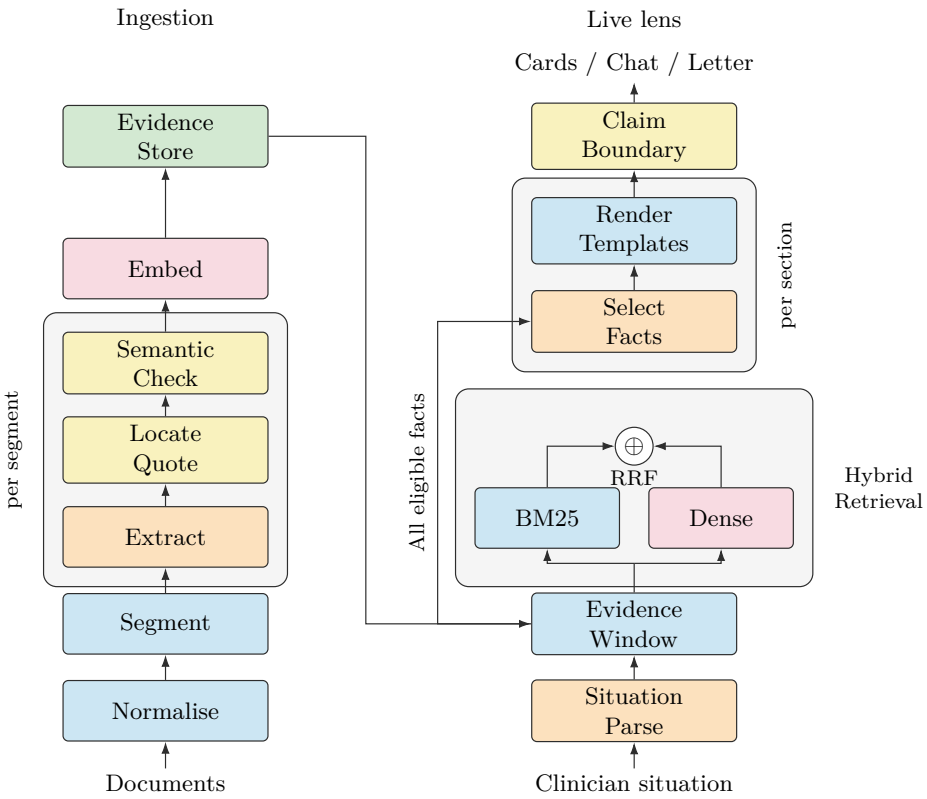


Figure 1: Ingestion and live lens construction. Orange denotes model calls, blue code, yellow checks, pink data or embeddings, and green storage throughout. Grey groups repeated or composite operations; $\oplus$ denotes RRF. Selection receives all eligible facts, while retrieval supports investigation and absence checks. Outputs cross the claim boundary.

(Google DeepMind, 2026), with pgvector storage (pgvector contributors, 2026). We fuse rank lists $\mathcal{R}$ with reciprocal rank fusion (Cormack et al., 2009):

$$\mathrm{RRF}(s) = \sum_{R \in \mathcal{R}: s \in R} \frac{1}{\kappa + \mathrm{rank}_R(s)}. \tag{2}$$

We restrict both branches to the same patient, revision and eligible documents before ranking. Duplicate segment identifiers contribute once per branch. A lexical fallback is recorded when embedding configuration or quota prevents the vector branch from running. A retrieval hit is a candidate passage, not a supported claim.

3.3 Lens Selection

We parse free-text situations into a role, task, concerns and as-of date. Configured situations use a fixed needs catalog; unfamiliar roles combine catalog entries. Selection returns eligible fact identifiers, completeness and an optional headline. It does not return arbitrary card prose. We render the selected facts through registered templates, run section selection in parallel and stream completed sections. A section title alone does not count as an answering card.

An empty selection does not prove that information is missing. We expand the concept with synonyms, check the eligible text lexically and run a full-text model pass over every eligible document. We show a documentation gap only when these checks succeed without finding evidence. A lexical failure, full-text failure or timeout produces an unavailable state. The statement is always relative to the searched record and as-of date; it is not a statement that the patient lacks the condition.

3.4 Claim Validation

Figure 2 shows the common claim boundary. Template claims require valid identifiers, located passages and supported fields. Composed claims additionally undergo a semantic comparison with the cited facts and source context. Absence claims require the completed documentation check. We keep clinician annotations and fixed nonclinical boilerplate separately typed.

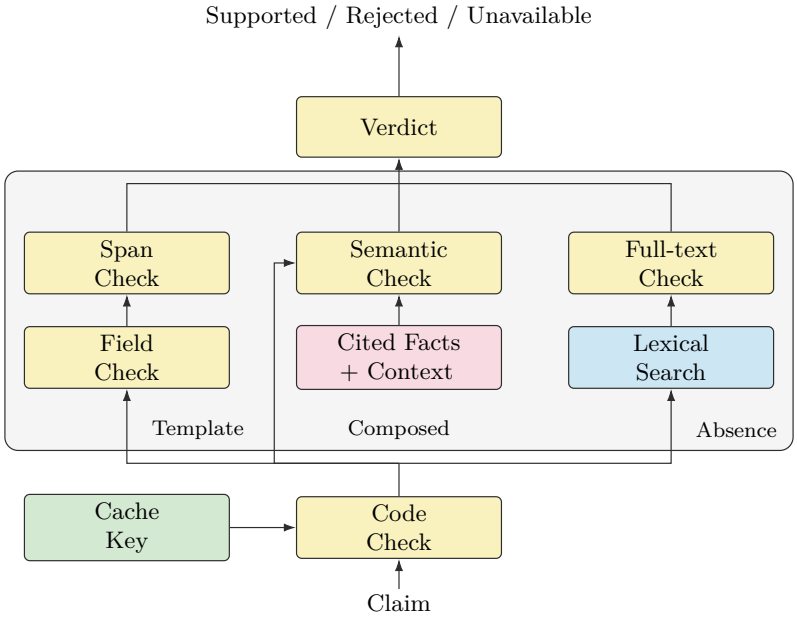


Figure 2: Claim boundary detail. Composed claims add a semantic check to code checks. Absence requires exhaustive lexical and full-text checks over the eligible record. The cache key binds the full evidence and claim identity. Check failures cannot establish absence.

We buffer generated answers until their sentence claims pass. A failed headline falls back to checked template lines. Letter edits retain citations and trigger revalidation; unsupported additions remain clinician

annotations. Text prepared for speech uses the validated answer. These paths share evidence identity and validation results rather than treating each output channel as a new source of truth.

We bind a cached validation to the patient, corpus revision, full evidence window, validator version, normalised claim text, cited fact identifiers and content hashes, and segment identifiers. An absence claim can change when an uncited document is added, which makes revision and window identity necessary even when the claim text and cited facts are unchanged. Review actions do not alter these machine verdicts.

3.5 Contradiction Adjudication

We propose candidates with rules for sides, dates, arithmetic and restated values, and with a model sweep over contextual evidence. We require evidence that the statements describe the same event or measurement before classifying a strict contradiction. Similar wording alone does not establish event identity.

We distinguish contradiction, need for clarification, change over time, measurement variation, omission and different context. Only contradictions and clarification requests reach the findings view. Medication changes between dated source lists are trends. Public titles and explanations pass claim validation, including when a candidate is downgraded to a clarification. We test expected findings and dismissal fixtures separately to measure false alarms as well as recovered cases.

4 Synthetic Data

We generate structured histories from seeded clinical scenarios, then plan which parts each document states, negates, paraphrases or omits. We render German documents from fresh templates and apply controlled mutations such as side flips, dose changes, date shifts and arithmetic inconsistencies. Figure 3 shows the pipeline. We assign patient and generation-family splits before optional rewriting and hold out styles, scenarios and mutation types.

We derive labels from the final rendered documents and the mutation ledger. A condition present in the hidden history but absent from every eligible document is not documented in the record. It must not become a supported claim merely because we know the generated history. We check that intended values, spans and mutations survive a prose rewrite; if a check fails, we retain the original rendering.

The saved manifest reports 100,000 generated patients, 900,210 documents and 4,824,477 code-checked facts. It records 60,963 contradictions, 470,690 distractors and 375,494 documentation gaps. These counts describe generated artifacts, not clinical prevalence, an ingested database cohort or independently verified semantic labels. Human approval of an aggregate count also does not make the underlying labels human gold. We follow the documentation goals of dataset datasheets (Gebru et al., 2021) and keep the annotation source and label tier explicit.

Bulk data, rendered pages and model artifacts remain local. Only a fresh synthetic sample is eligible for repository distribution. We do not use challenge prose in templates, training examples, publication examples or source-controlled test fixtures. A separate human-reviewed sample is needed to estimate label noise and clinical plausibility.

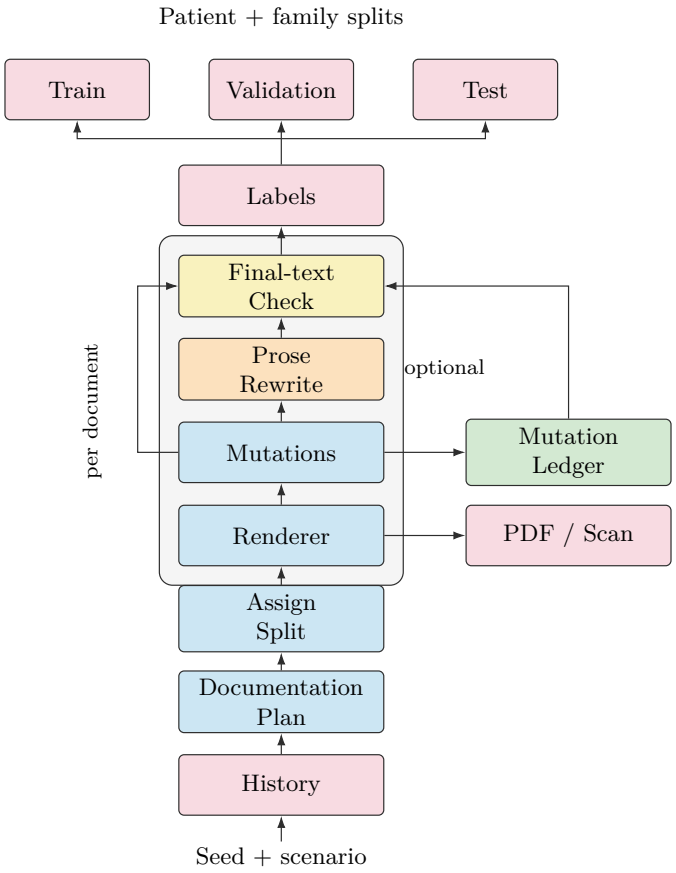


Figure 3: Synthetic data construction. Patient and family splits precede rewriting. Labels follow the final text and mutation ledger. Optional rewrites retain the original on check failure. Code-checked labels are distinct from human gold.

5 Verifier Training

We formulate verification as classification of a claim and a passage with context into supported, unsupported or contradicted. We construct hard negatives from controlled mutations and keep assertion and timing information. We split by patient and family before adaptation. The training, validation and test manifests contain 24,000, 3,000 and 3,000 examples, respectively. We keep real-patient weak labels outside training and report them separately.

We adapt a quantised Qwen3 base (Yang et al., 2025) with low-rank updates (Hu et al., 2021) using MLX on Apple silicon (Hannun et al., 2023). For a frozen weight matrix W_0, the adapted projection has the form

$$h = W_0 x + \frac{\alpha}{r} BAx, \qquad A \in \mathbb{R}^{r \times d}, \quad B \in \mathbb{R}^{m \times r}. \tag{3}$$

The quantised-base approach is related to QLoRA (Dettmers et al., 2023). We do not equate this MLX run with the original CUDA implementation or assert that it uses all of its quantisation methods. We mask the prompt during training and bound the input length at 512 tokens. The configured wall-time cap is 45 minutes. The full-data run reached that cap before completing its epoch. We evaluate the saved checkpoint, rather than claiming completed training.

We compare rules-only checks, the configured proxy semantic judge and the local adapter. We measure macro F-score, coverage and latency together. We reserve model confidence and calibration for experiments with recorded probabilities; a class prediction alone does not define a reliability curve. Calibration analysis follows the distinction between confidence and accuracy made by Guo et al. (2017). We use a model card to record the training conditions and intended use (Mitchell et al., 2019). The proxy judge remains the default semantic backend unless a separate integration decision approves the local verifier.

6 Results

6.1 Evaluation Protocol

We keep the challenge regression, synthetic pipeline, retrieval and verifier experiments separate. Evaluation uses an empty review layer. The primary challenge review covers 15 answers across 5 situations, entered once as free text and once through configured chips. We use existing review records and do not run or publish new challenge blind results.

Every empirical value enters through `generated/numbers.tex`. We extract numeric aggregates from allowlisted result fields and retain source hashes. The measurement review binds exact values to those hashes. The reviewer is an agent acting under the author's delegation, not a clinician. Synthetic labels remain code checked. We omit unmeasured browser visibility, blind relevance and vector-only retrieval results rather than treating them as observations.

6.2 Challenge Regression

Table 1 reports the primary cold API matrices and the source audit. Each cold matrix answered 15/15 questions. We located 166/166 cited facts and found 0 unsupported lines among 167 distinct reviewed card lines. We recovered 5 qualifying contradictions with 0 false findings. These checks cover API screen data; they do not certify browser visibility, chat, speech, letters or clinical correctness. The slowest primary cold

Measure	Observed
Primary cold answers (text and chip, each)	15/15
Cited facts located / checked	166/166
Unsupported / distinct card lines reviewed	0/167
True contradictions / false findings	5/0
Slowest primary cold API completion (s)	10.138
Slowest confirmed cached API completion (s)	0.239

Table 1: Agent-reviewed challenge API checks. Counts retain their own denominators; the located-fact audit also includes repeat and cached requests.

completion, 10.138 s, misses the under-10 s challenge target. Confirmed cache hits completed within 0.239 s. Additional cache-fill requests failed availability or strict source and warning checks. We therefore do not

claim that every repeated or cached run passed. The older saved automatic regression reports 12/15 answer passes and 3/15 full-suite passes; it is a separate run and does not replace the primary review.

6.3 Synthetic Pipeline and Retrieval

Table 2 separates generated data from executed evaluation cases. We do not count an unexecuted challenge as a pass or a failure. The number of generated patients is also distinct from the number ingested into the shared database. Table 3 compares query-level retrieval aggregates at the configured cutoffs. These experiments concern synthetic query labels and do not establish challenge-patient recall. Figure 4 shows measured verifier reliability and the latency/quality comparison.

Research artifact	Observed
Generated synthetic patients	100,000
Generated synthetic documents	900,210
Code-checked facts / contradictions	4,824,477/ 60,963
Distractors / documentation gaps	470,690/ 375,494
Patients with rewritten prose	300
Sample patients / documents	20/ 190
Synthetic challenges generated	150
Synthetic challenges evaluated	27
Verifier training / validation / test examples	24,000/ 3,000/ 3,000
Verifier macro F-score	0.764
Verifier accuracy	0.787
Verifier median latency (ms)	629.132
Retrieval recall at configured cutoff	0.371

Table 2: Synthetic and research artifact counts. Code-checked labels remain distinct from human gold. An experiment can have generated cases without measured cases.

Retrieval	Queries	Recall@5	Recall@10	MRR@10
BM25	32	0.362	0.371	0.344
Hybrid	32	0.362	0.371	0.344

Table 3: Synthetic retrieval evaluation. We omit the unmeasured vector-only mode; the identical BM25 and hybrid scores do not establish a vector retrieval benefit.

6.4 Verifier Comparison

Table 4 reports each method's own sample count. Comparisons require matching evaluation populations and label definitions; unequal counts preclude a paired interpretation. Macro F-score averages class-specific F-scores, and coverage records whether the method produced a decision. We do not infer calibrated confidence from classification accuracy or treat rules-only agreement as human annotation. On the synthetic test set,

Verifier	Examples	Macro F-score	Coverage	Median ms
Rules only	300	0.295	1	0.068
Proxy semantic	300	0.896	1	1739.708
Local adapter	300	0.764	1	629.132
Local + GPT cascade	300	0.956	1	785.210

Table 4: Verifier results on synthetic held-out generation families, kept separate from held-out mutation experiments and real-patient weak labels. Median latency is specific to the saved evaluation conditions.

the local verifier reaches 0.764 macro F-score, below GPT at 0.896 and above rules at 0.295. A validation-selected cascade reaches 0.956 and routes a projected 55% of claims away from GPT. This is offline replay

with both predictions collected; it is not a measurement of reduced live service traffic. The real-text set, labelled `real-gold` in the result file, contains rule-derived test labels rather than clinician gold. On its 100 claims, the cascade scores 0.616 versus GPT's 0.657 and the local model's 0.421. Macro F-score includes absent classes, which limits comparisons with the balanced synthetic task.

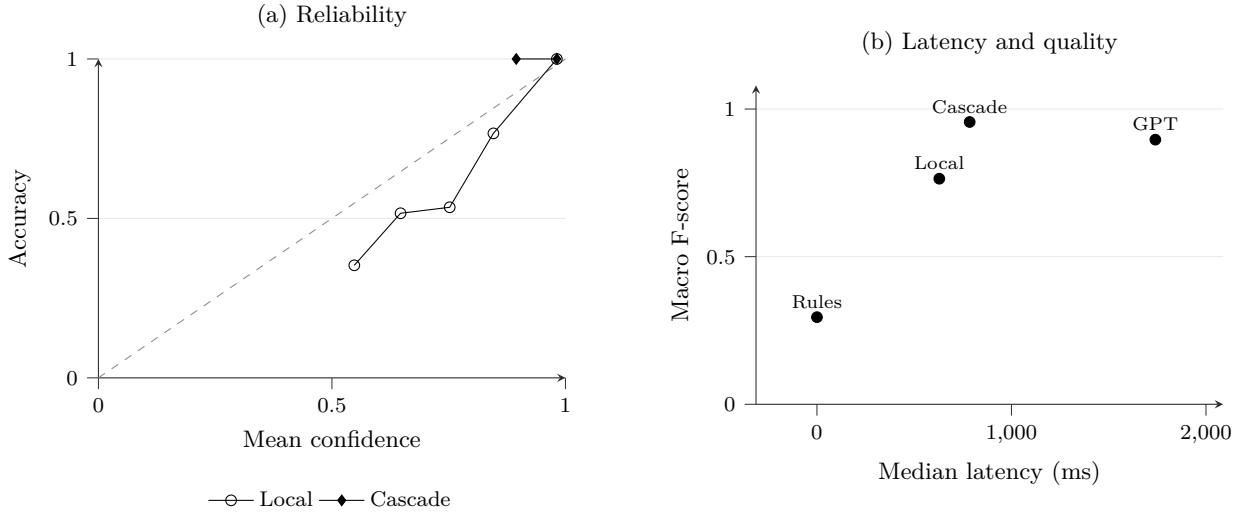


Figure 4: Synthetic test-set aggregates from the same evaluation. Reliability uses calibrated local scores and the accepted local subset of the cascade; GPT and rules provide no measured confidence. The dashed line denotes perfect calibration. Latency includes the saved request times, with projected serial routing for the cascade.

6.5 Limitations

We evaluate a narrow challenge workflow and generated records with code-checked labels. The primary review is agent performed, the cached checks expose source and warning failures, and cold completion misses the challenge target. Located quotes cannot establish semantic correctness, translation fidelity or complete selection. The local verifier and cascade trail GPT on real text; the real-text set is test only and its rule-derived labels do not provide clinician gold. PDF/scan recall, independent relevance annotation and browser completion are outside our measured claims. The hosted demonstration is public without authentication. We establish neither clinical efficacy nor reduced workload, and prospective deployment requires independent annotation, workflow evaluation and access-control review.

7 Conclusion

We construct task-specific clinical views by separating located evidence, fact selection, template rendering and a shared claim boundary. The primary challenge API review answers 15/15 questions in each cold matrix, but its 10.138 s maximum misses the latency target. Our synthetic cascade reaches 0.956 macro F-score with 55% projected GPT-call savings; its real-text score of 0.616 remains below GPT's 0.657. We retain GPT as the application's semantic verifier. Our separate IBM hardware experiment characterises a small synthetic kernel block and makes no claim of clinical or quantum advantage.

A Exploratory Quantum Experiment

We use IBM Qiskit for a synthetic-only experiment outside the clinical runtime. We balance supported claims against unsupported and contradicted claims, using 400 training and 200 test examples from frozen verifier splits. The binary task differs from the main verifier's label space. Pair features combine cosine similarity, products and absolute differences of character-hash vectors. We fit PCA and scaling on training features and encode 8 components in a ZZ feature map with 2 repetitions. The intended Gemini feature experiment was blocked by quota, so we report the offline character-hash control and do not infer semantic-embedding

performance.

On `ibm_marrakesh`, we execute 136 upper-triangular overlap circuits, including diagonals, for a 16 by 16 test-kernel block. We use 1,024 shots per circuit. Median transpiled depth is 760 and recorded execution time is 52.903 s. The mean absolute error relative to the exact simulator block is 0.066. These measurements describe kernel fidelity, not classification accuracy. A test-to-test block cannot replace the classifier's test-to-training kernel, so we make no held-out hardware classifier claim. The full aggregate and backend provenance remain in `ml/quantum/results.json`.

We also implement QAOA evidence selection on small synthetic candidate sets, compared with exact and greedy search. This is a relevance surrogate rather than a clinical standard, and exact enumeration is practical at this scale. The experiments establish neither scaling behaviour, quantum advantage nor transfer to patient care.

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